

Specifications

ASIC for optical sensor

S E P 1 1

SPEC NO. 1209-1
ISSUE 21/SEP./2012

CUSTOMERS APPROVAL

DATE: _____

BY: _____

PRESENTED

DATE: _____

PREPARE _____

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■ Concerning the handling of this specifications.

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■ Concerning the handling of this product.

In case of using this product, please observe application conditions of this specifications and following attentions.

We do not take all responsibility about the damage that occurred by having used this product without observing application conditions of this specifications and following attentions.

【Attentions】

(1) This product was made for the purpose of the following use as a general rule.

In addition, even if it is following use, please observe the attention (2), in case that this product is used in various safety devices of mention to (2)

And even if it is following use, do not use this product in case that the instruments of mention to (3) consists of this product

- OA equipment • Measuring instruments and apparatus • Machine tool
- Audio visual system • Household electric appliance
- telecommunications equipment (Without main line)

(2) When this product is used in the following apparatus that particularly high reliability is required, please contact us beforehand by all means, and please take the appropriate measures such as a fail-safe design or the redundant design of the apparatus side in the responsibility of the customer.

- Unit about control or various safety of the transportation equipment (vehicles for locomotion by land, air, water or rail etc.)
- large-sized computer • Traffic control signal • Gas leak detection crossing gate
- Disaster security equipment • Other various safety devices, etc.

③ Please do not use this product for the following apparatuses that extremely high reliability is required in a function or precision.

- Aerospace instrument • Telecommunication equipment (main line)
- Nuclear power control equipment • Medical life support equipment

(4) When there is a doubt which of above (1)(2)(3) your apparatuses correspond to, please confirm in us.

■ If you have any questions, please contact us about this product beforehand.

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1. Description

SEP11 is specifically designed for optical sensors with half-dividing photodiodes loaded.

2. Feature

- ① Built-in half-dividing photodiode (0.75mm×0.55mm×2 pieces)
- ② Built-in sensitivity adjustment function
- ③ Built-in output overcurrent detection function
- ④ Sync/Async mode selection function
- ⑤ RoHS(*) compliance

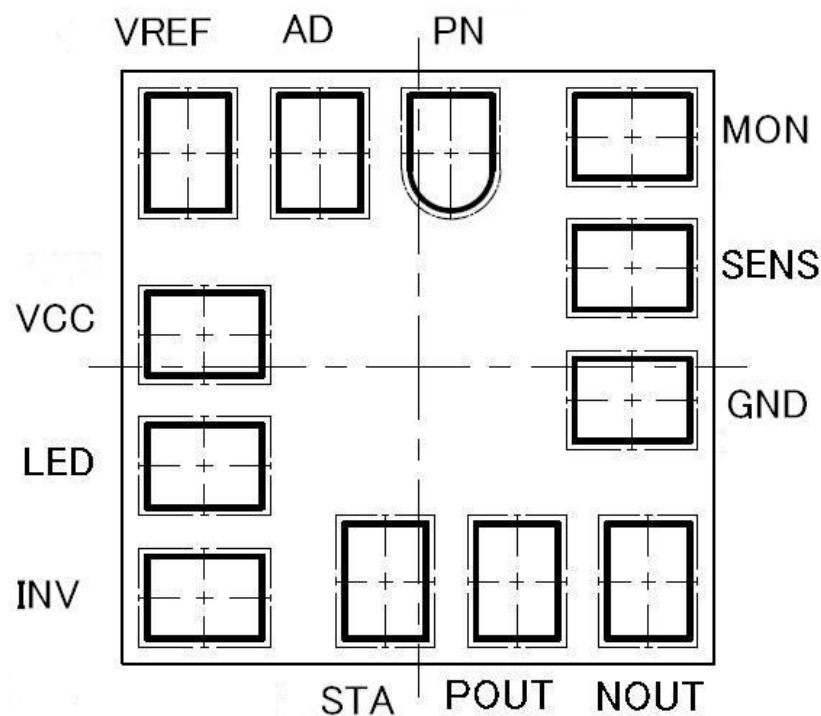
(*)Restriction of the use of certain Hazardous Substances in electrical and electronic equipment

3. Basic characteristics

Structure	: Silicon monolithic
Wafer	: P-type silicon wafer
Process	: Bi-CMOS
Package	: 12 pins COB
Use	: Optical sensors

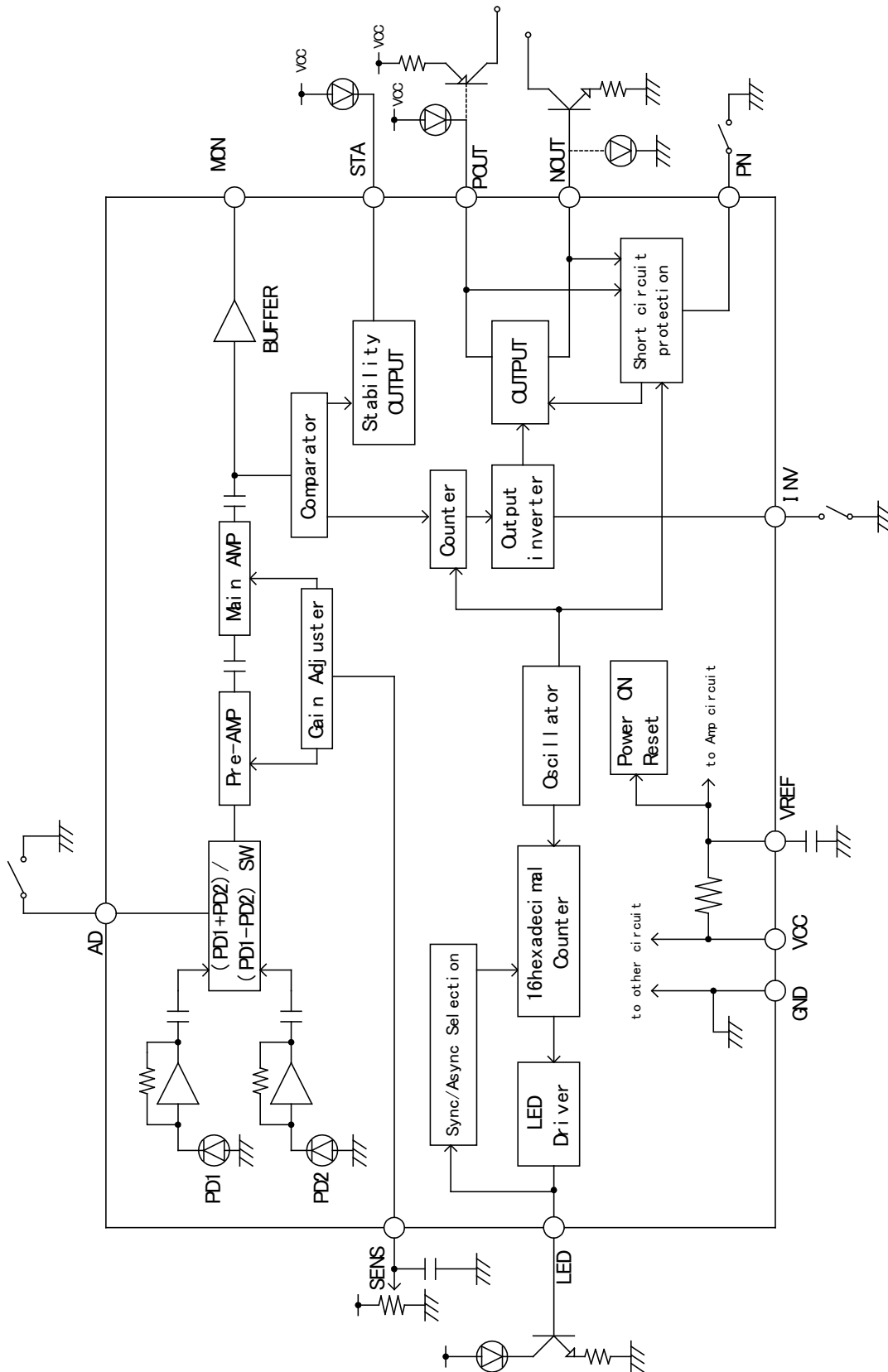
This ASIC is not designed for Radiation endurance.

4. Pin configuration (Bottom View)



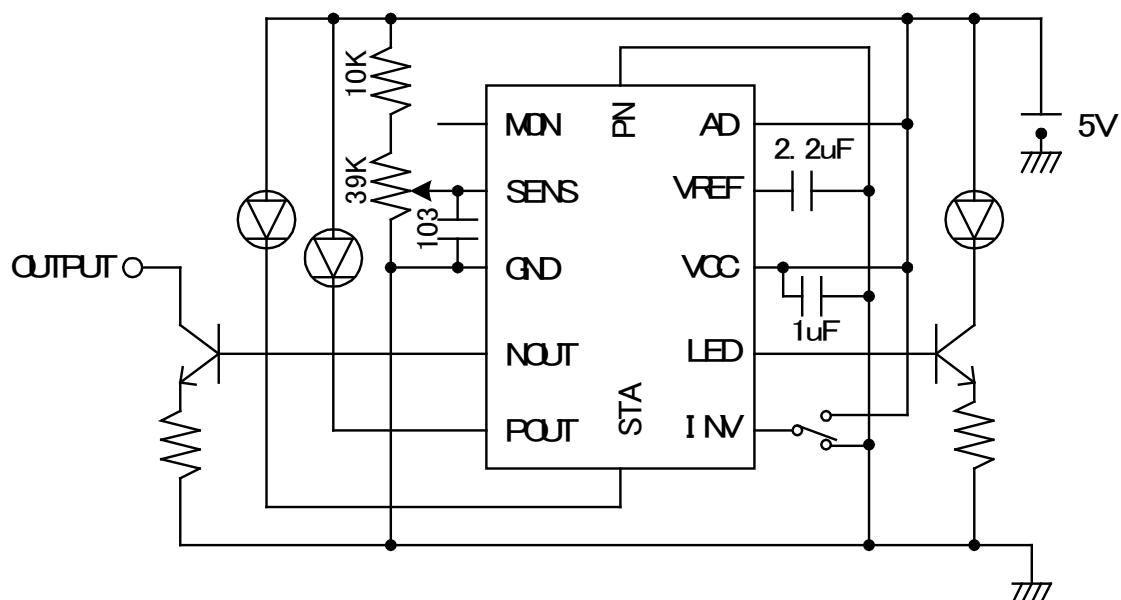
(Bottom View)

5. Block diagram

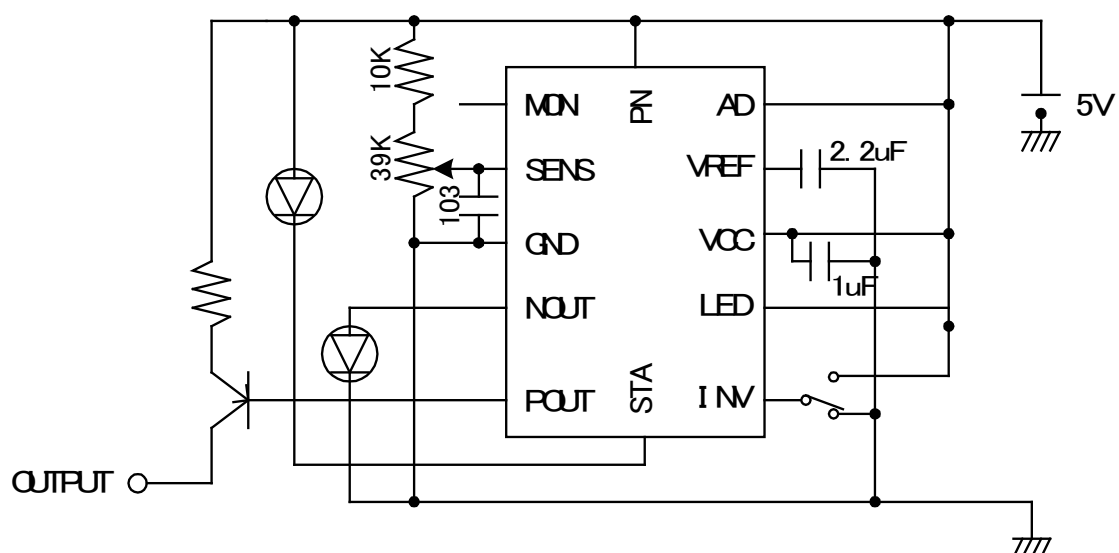


6 . Basic application

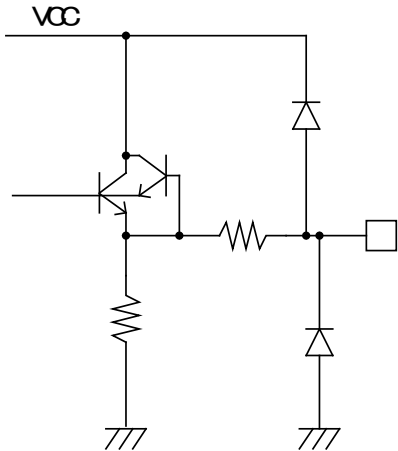
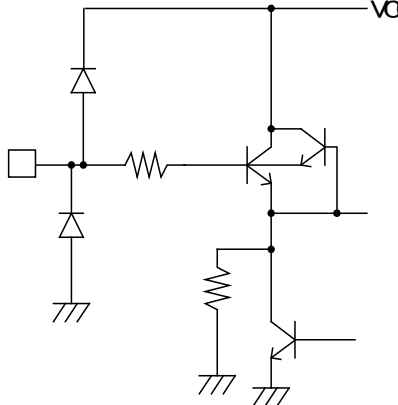
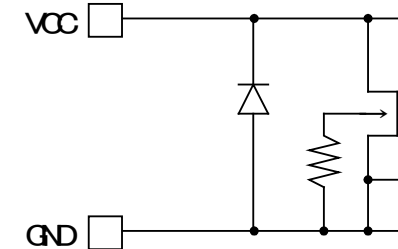
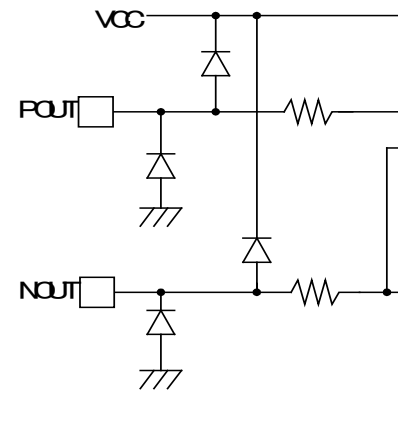
Synchronous , NPN Out put

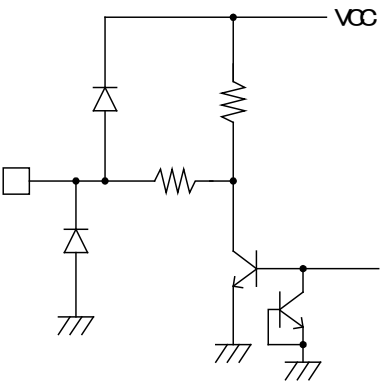
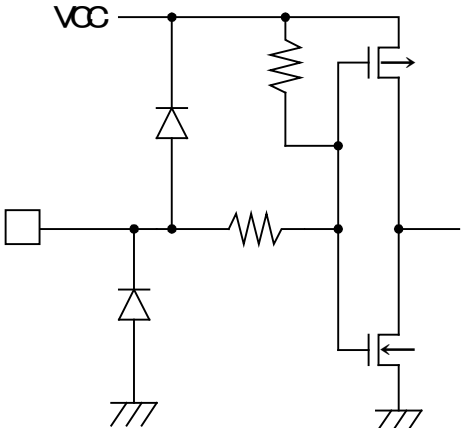
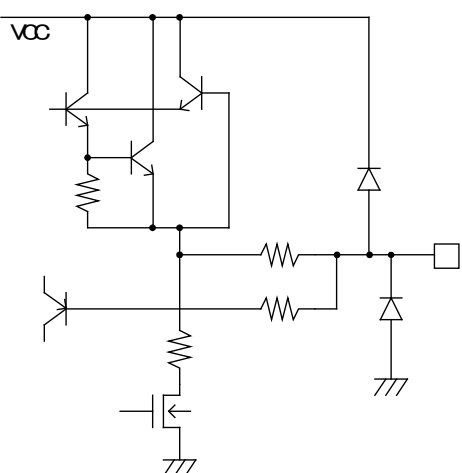


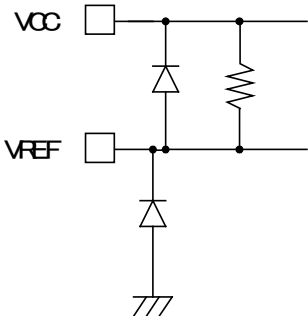
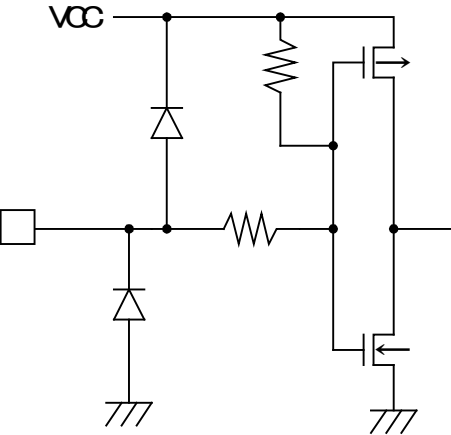
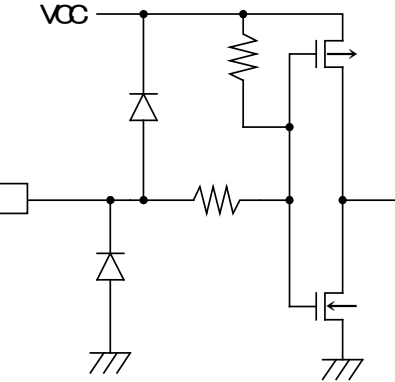
Asynchronous , PNP Out put



7. The outskirts circuit of each terminals and terminal explanation

NO.	PIN name	The outskirts circuit of terminals	Pin explanation
1	MON		Check Pin for amplifier output signal.
2	SENS		This pin function is the gain adjustment of the amplifier for sensitivity adjustment. This sensitivity can be adjusted in proportion to the square of the voltage of this pin.
3 9	GND VCC		These are power supply pin and ground pin. Please connect VCC pin with GND pin by a capacitor by all means.
4 5	NOUT POUT		These pins output whether light signals are received . When these are turned on, the constant current is carried into POUT pin, and it is carried off from NOUT pin. And these output pins contain overcurrent detection circuits each other. Either overcurrent detection circuit is active according to PN pin condition.

NO.	PIN name	The outskirts circuit of terminals	Pin explanation
6	STA		This is an output terminal for the stability indication. When the signal is beyond stability light level or under stability shadow level, STA pin turns ON.
7	INV		In case of INV pin being HI level or opening, NOUT and POUT are turned on when the light signal is received. In case of INV pin being LOW level, NOUT and POUT are turned on when the light signal is not received.
8	LED		This is a terminal outputting a pulse for LED drive. In case of asynchronous mode, please connect this terminal to VCC.

NO.	PIN name	The outskirts circuit of terminals	Pin explanation
1 0	VREF		This is a power supply terminal for IV conversion circuit and amplifier block. Please connect VREF pin with GND pin by a capacitor by all means.
1 1	AD		According to this pin condition, it is decided whether light receiving signals from half-dividing photodiodes are added or are subtracted from. In case of AD pin being HI level or opening, these signals are added. In case of AD pin being LOW level, these signals are subtracted from.
1 2	PN		In case of PN pin being HI level or opening, the overcurrent detection circuit of POUT is active. In case of PN pin being LOW level, the overcurrent detection circuit of NOUT is active.

8. Function explanation

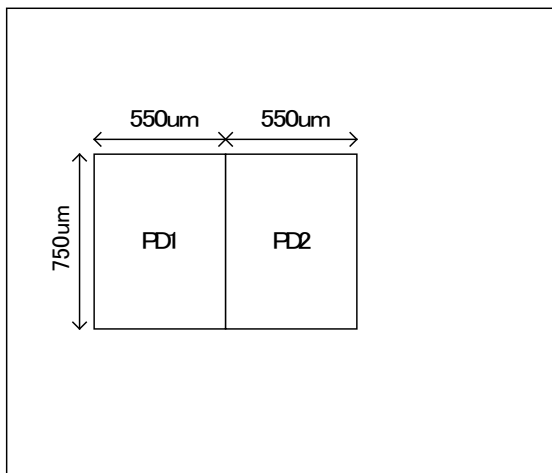
(IV conversion circuit)

The light current from a photodiode is converted into the voltage by a conversion resistance $44\text{k}\Omega$ (TYP). This conversion resistance has a line graph characteristics which decreases when light currents increase

(Half-dividing photodiodes)

This ASIC has a half-dividing photodiodes that are PD1 and PD2. And they are connected with each IV conversion circuit. The each current from PD1 and PD2 are converted into the each voltage (V_{pd1} and V_{pd2}).

In case of AD pin being HI level or opening, these voltages are added ($V_{pd1}+V_{pd2}$). In case of AD pin being LOW level, V_{pd2} are subtracted from V_{pd1} ($V_{pd1}-V_{pd2}$). And it is led to amplifier circuit.



(Amplifier circuit)

The amplifier circuits consist of the pre-amp and the main amp, and the total gain of amplifiers is about 57dB(TYP). Each amplifier has a gain adjustment circuit which gain changes with the voltage given from a SENS pin at the same time. By this function, the total gain can be changed while maintaining relation of the square for the SENS pin voltage.

(Levels for comparator circuit)

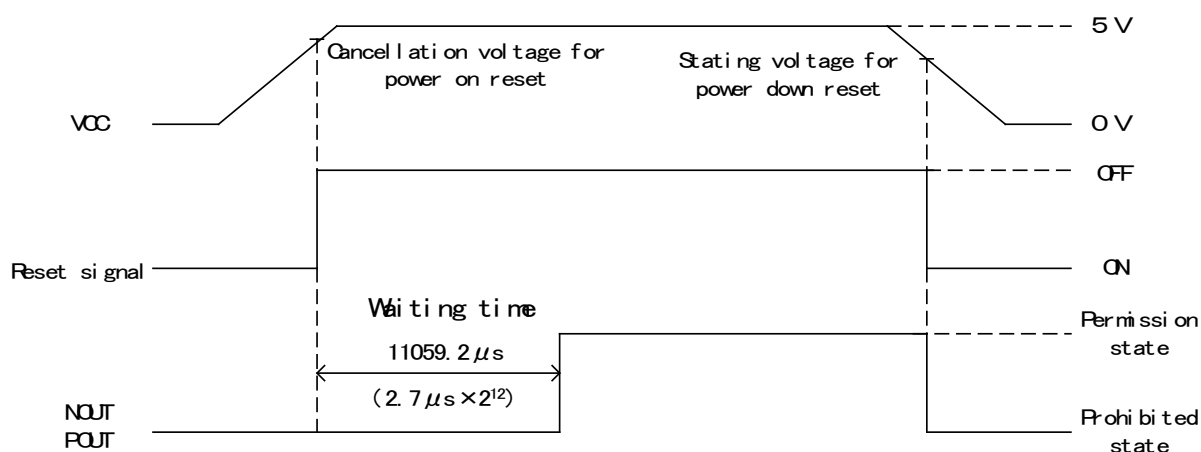
Level name	Level		Explanation
Stable light	110%		When the signal is beyond this level, STA pin turns ON.
Light	100%		When the signal is beyond this level consecutively eight times, NOUT and POUT turn on (or off).
Shadow	SENS MAX	SENS MIN	When the signal is under this level consecutively eight times, NOUT and POUT turn off (or on).
	75%	93%	
Stable shadow	SENS MAX	SENS MIN	When the signal is under this level, STA pin turns ON.
	58%	77%	

The shadow level and the stable shadow level are changed continually by SENS pin voltage.

(Power on reset function and waiting time)

When the VCC pin voltage is beyond the cancellation voltage for power on reset, reset state of logic circuit is canceled and all function are active. But the NOUT pin and POUT pin continue prohibited state during about 11msec period.

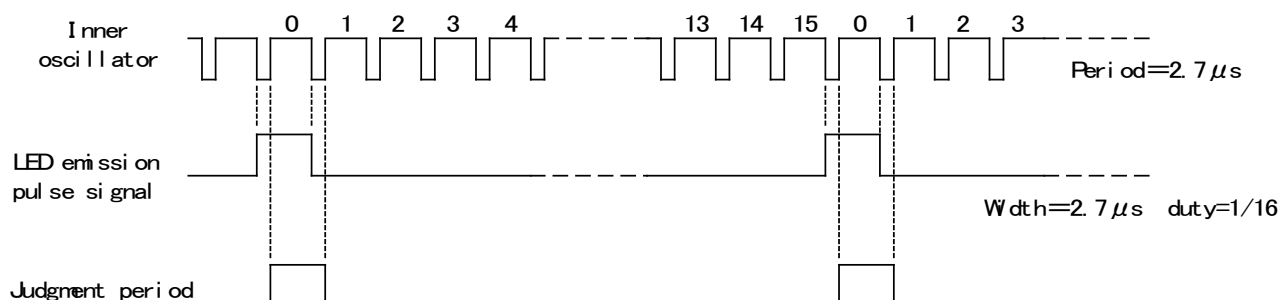
When the VCC pin voltage is under the starting voltage for power down reset, all logic circuit become reset state and NOUT pin and POUT pin become prohibited state.



(LED emission timing and judgment period in case of a synchronous mode)

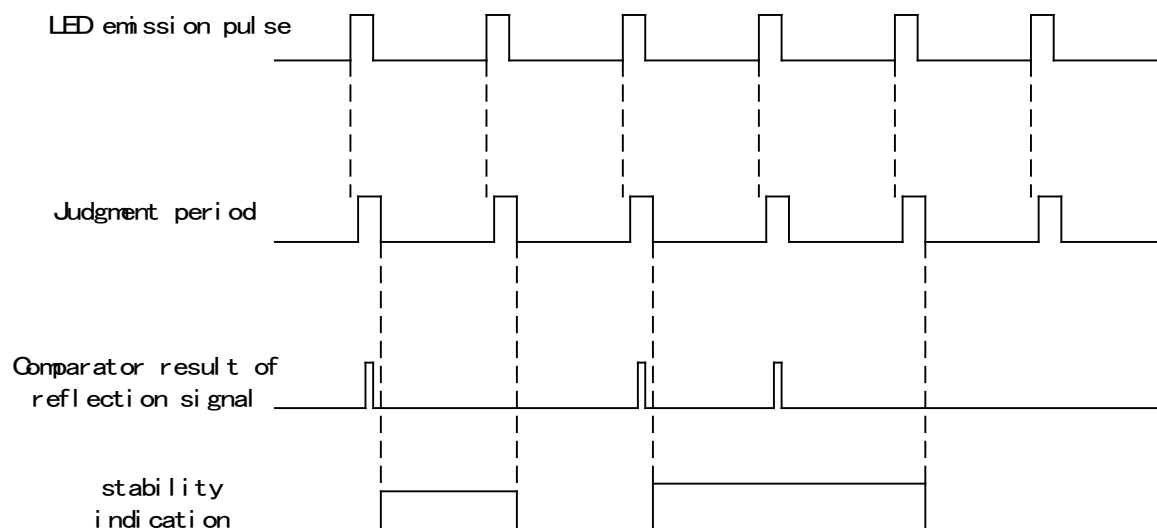
The LED emission pulse signal (width=2.7us(TYP), period=2.7us × 16=43.2us(TYP)) is created by inner pulse oscillator (period=2.7us(TYP) duty=1/5). And it is output for the external transistor that drives LED.

The judgment period of a reflection signal delay during 1/5 period from the LED emission pulse signal because the reflection signal delay by IV conversion circuit and amplifier.



(Stability indication)

When the reflection signal in the judgment period is beyond the stable light level, or it is under the stable shadow level, stability indication (STA) turns on. The state of stability indication continues until the end of next judgment period.

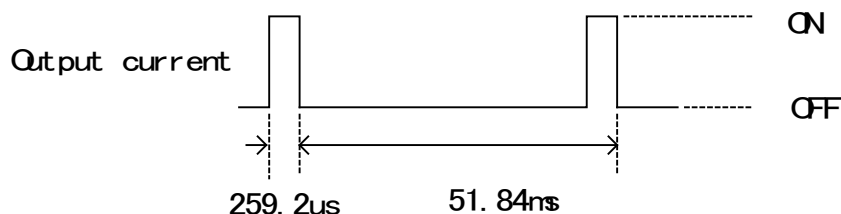


(Overcurrent detection circuit of output pins)

When the output (output = NOUT pin or POUT pin) voltage continues beyond the overcurrent detection voltage during more than 259.2us(TYP), it is judged that the overcurrent is carried into an external output transistor. And output currents are turned off between 51.84ms(TYP).

Though output current turned on more afterward, when the output voltage continues beyond the overcurrent detection voltage during more than 259.2us(TYP) again, output current are also turned off.

If the overcurrent is carried into the external output transistor, the output pin works as noted above.



(An asynchronous mode)

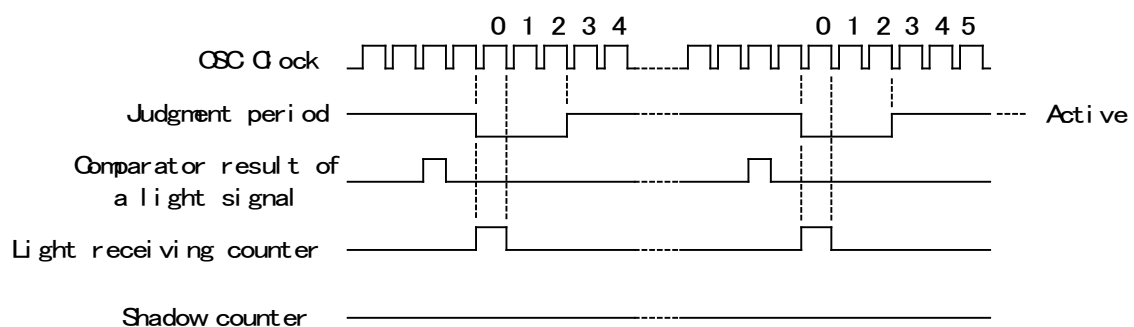
When the LED pin is connected to VCC pin, this ASIC works as an asynchronous mode.

This is the operating mode when this ASIC receives the asynchronous light.

A judgment period is activated until the light signal is received. And it stops 1clock late after the light signal is received. And it is activated 3clocks late after it stops.

When the light signal is received between 11clocks, a judgment period is also stops 1clock late and a light receiving counter is counted.

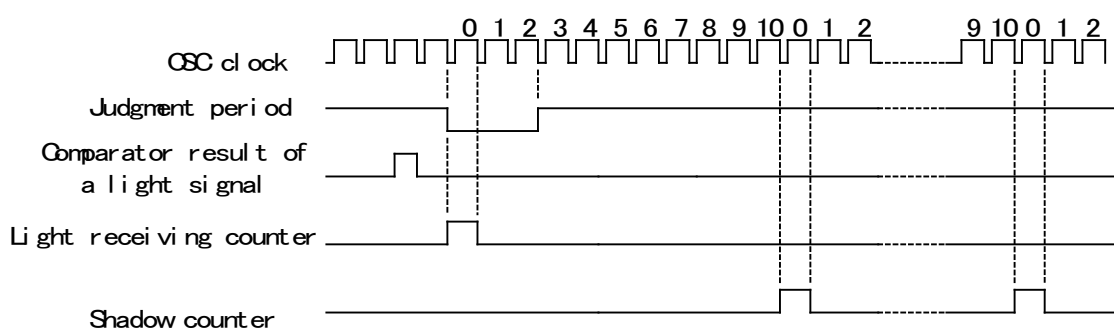
If the light receiving counter is counted consecutively eight times, this ASIC judges that the light is received, and NOUT pin and POUT pin are turned on.



A judgment period is activated until the light signal is received. And it stops 1clock late after the light signal is received. And it is activated 3clocks late after it stops. When the light signal is not received between 11clocks, a shadow counter is counted.

Afterward, when the light signal is not received all the time, a judgment period is activated all time and the shadow counter is counted every 11clocks.

If the shadow counter is counted consecutively eight times, it is judged that the light is not received, and NOUT pin and POUT pin are turned off



Besides, in case of an asynchronous mode, this ASIC assumes the light pulse signal of the following conditions.

	MIN	TYP	MAX	UNIT
Pulse width	—	2.7	—	us
Pulse period	18	20	22	us

9. Attention

- ① Please do not make GND pin opening, but connect GND pin with the minimum potential to be supplied to this ASIC by all means
- ② Please connect a decoupling capacitor between VCC pin and GND pin, and between VREF pin and GND pin by all means.
- ③ Please connect a decoupling capacitor between SENS pin and GND pin by all means. The value of this capacitor is less than 0.01uF in case the value of resistance between SENS pin and GND pin is 50kΩ. In case of the other value, Please confirm that the voltage of SENS pin is fixed within 5ms after the power supply voltage turns on. If the voltage of SENS pin is fixed later, the output terminal can do error operation.
- ④ Because this IC has high sensitivity amplifiers built-in and is easy to be affected by a noise, it is recommended that it is used in shield condition.
- ⑤ Please design the board pattern of the MON pin as short and small as possible not to be affected by a noise.

10. Absolute Maximum Ratings

Parameter	Symbol	Condition	Rating			Unit
			MIN	TYP	MIN	
Supply voltage	VCCM		−0.3		6	V
Input pin voltage	VINM		−0.3		VCC+0.3	V
Output pin voltage	VOM		−0.3		VCC+0.3	V
Operating temperature range	Topr		−20		65	°C
Storage temperature range	Tstg		−30		125	°C
Power dissipation		Ta ≤ 25°C	—	463	—	mW
Reduction rate		Ta > 25°C	—	4.63	—	mW/°C

【Recommended Condition】

Parameter	Symbol	Condition	Rating			Unit
			MIN	TYP	MIN	
Supply voltage range	VCC		4.5	5.0	5.5	V
Pulse width of light signal	tTW	An asynchronous mode	—	2.7	—	us
Pulse period of light signal	tTP		18	20	22	us

11. Electrical Characteristics

【Parameter to be inspected prior to shipment】 VCC=5V, Ta=25°C

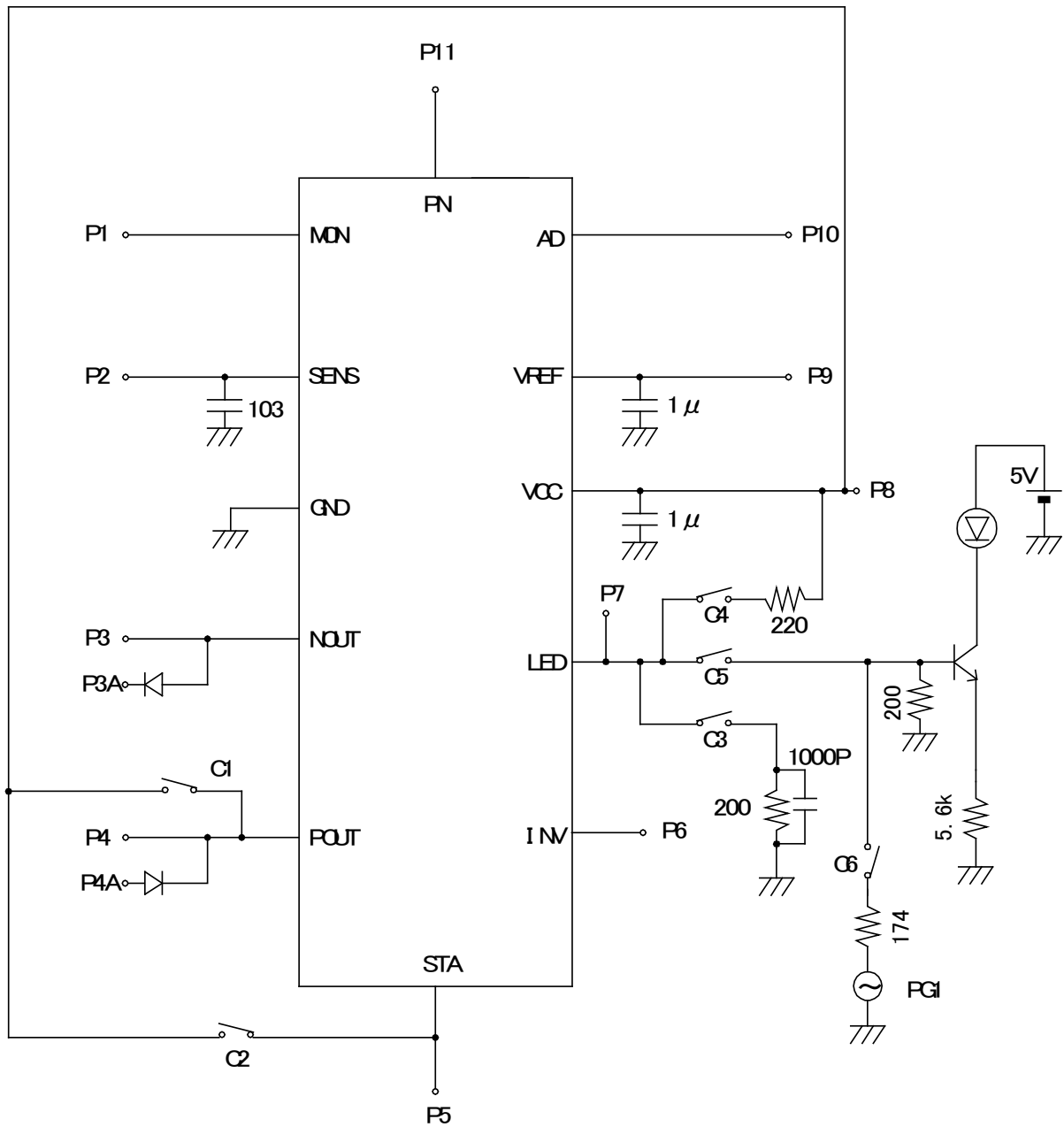
NO.	Parameter	Symbol	Measuer pin	Condition	SPEC			
					MIN	TYP	MAX	UNIT
1	Cancellation voltage for power on reset	VRSP	P8	Move P8 voltage from 3V to 5V. Measure P8 voltage when P3 current is carried off more	3.8	4	4.2	V
2	Starting voltage for power down reset	VRST	P8	Move P8 voltage from 5V to 3V. Measure P8 voltage when P3 current is carried off less	3.4	3.6	3.8	V
3	Supply current (output OFF)	ICC1	P8	Measure the current carried into P8.	4.9	6.3	7.7	mA
4	Supply current (output ON)	ICC2	P8	Measure the current carried into P8.	6.5	8.4	10.3	mA
5	NOUT output current	INOM	P3	Measure the current carried off from P3.	-3.42	-2.5	-1.86	mA
6	POUT output current	IPOM	P4	Measure the current carried into P4.	1.69	2.3	3.11	mA
7	STA output current	ISTM	P5	Measure the current carried into P5.	2.4	3.6	4.3	mA
8	VREF pin voltage	VREF	P9	Measure the voltage of P9.	4.6	4.7	4.9	V
9	LED emission pulse period	TLS	P7	Measure the pulse period at P7.	32.8	43	52.8	us
10	LED emission pulse width	TLP	P7	Measure the pulse width at P7.	2.05	2.7	3.3	us
11	Overcurrent detection time(NOUT)	TSSN	P3	Measure the pulse width at P3.	196.8	259.2	316.8	us
12	Overcurrent detection period(NOUT)	TSPN	P3	Measure the pulse period at P3.	39.36	51.84	63.36	ms
13	Overcurrent detection time(POUT)	TSSP	P4	Measure the pulse width at P4.	196.8	259.2	316.8	us
14	Overcurrent detection period(POUT)	TSPP	P4	Measure the pulse period at P4.	39.36	51.84	63.36	ms
15	Overcurrent detection voltage(NOUT)	VSN	P3	Move P3A voltage from 0.3V to 0.9V. Measure the P3 voltage when a pulse at P4 starts.	1.1	1.3	1.5	V
16	Overcurrent detection voltage(POUT)	VSP	P4	Move P4A voltage from 4.7V to 4.1V. Measure the P4 voltage when a pulse at P3 starts.	3.5	3.7	3.9	V
17	Movement test of a synchronous mode	IH	P3		-3.6	-2.5	-1.6	mA
18	Movement test of an asynchronous mode	IT	P3		-3.6	-2.5	-1.6	mA

Conditions to be inspected prior to shipment

NO.	Parameter	SENS	NOUT	NOUT	POUT	POUT	STA	INV	VCC	AD	PN		POUT	STA	LOAD	ASYN	SYN	ASYN SIG
		P1	P3	P3A	P4	P4A	P5	P6	P8	P10	P11	PG1	C1	C2	C3	C4	C5	C6
1	Cancellation voltage for power on reset	0.0	0.0	—	—	—	—	0.0	3→5	0.0	0.0	—	ON	ON	OFF	OFF	OFF	OFF
2	Starting voltage for power down reset	0.0	0.0	—	—	—	—	0.0	5→3	0.0	0.0	—	ON	ON	↓	↓	↓	↓
3	Supply current (output OFF)	0.0	0.0	—	—	—	—	3.5	5.0	0.0	0.0	—	ON	ON	↓	↓	↓	↓
4	Supply current (output ON)	0.0	0.0	—	—	—	—	1.5	5.0	0.0	0.0	—	ON	ON	↓	↓	↓	↓
5	NOUT output current	0.0	0.0	—	5.0	—	5.0	0.0	5.0	0.0	0.0	—	OFF	OFF	↓	↓	↓	↓
6	POUT output current	0.0	0.0	—	5.0	—	5.0	0.0	5.0	0.0	0.0	—	↓	↓	↓	↓	↓	↓
7	STA output current	0.0	0.0	—	5.0	—	5.0	0.0	5.0	0.0	0.0	—	↓	↓	↓	↓	↓	↓
8	VREF pin voltage	0.0	0.0	—	5.0	—	5.0	0.0	5.0	0.0	0.0	—	↓	↓	↓	↓	↓	↓
9	LED emission pulse period	0.0	0.0	—	5.0	—	5.0	0.0	5.0	0.0	0.0	—	↓	↓	ON	↓	↓	↓
10	LED emission pulse width	0.0	0.0	—	5.0	—	5.0	0.0	5.0	0.0	0.0	—	↓	↓	OFF	↓	↓	↓
11	Overcurrent detection time(NOUT)	0.0	—	0.9	—	5.0	5.0	0.0	5.0	0.0	1.5	—	↓	↓	↓	↓	↓	↓
12	Overcurrent detection period(NOUT)	0.0	—	0.9	—	5.0	5.0	0.0	5.0	0.0	1.5	—	↓	↓	↓	↓	↓	↓
13	Overcurrent detection time(POUT)	0.0	—	0.0	—	4.1	5.0	0.0	5.0	0.0	—	—	↓	↓	↓	↓	↓	↓
14	Overcurrent detection period(POUT)	0.0	—	0.0	—	4.1	5.0	0.0	5.0	0.0	3.5	—	↓	↓	↓	↓	↓	↓
15	Overcurrent detection voltage(NOUT)	0.0	—	0.3→0.9	—	5.0	5.0	0.0	5.0	0.0	0.0	—	↓	↓	↓	↓	↓	↓
16	Overcurrent detection voltage(POUT)	0.0	—	0.0	—	4.7→4.1	5.0	0.0	5.0	0.0	5.0	—	↓	↓	↓	↓	↓	↓
17	Movement test of a synchronous mode	5.0	0.0	—	5.0	—	5.0	3.5	5.0	0.0	—	—	↓	↓	↓	↓	ON	↓
18	Movement test of an asynchronous mode	5.0	0.0	—	5.0	—	5.0	3.5	5.0	0.0	—	W1	↓	↓	↓	ON	OFF	ON

W1: Pulse width=2.7us, period=20us, pulse amplitude=4.3V

Testing circuit to be inspected prior to shipment



【Parameter of wafer testing】 VCC=5V, Ta=25℃

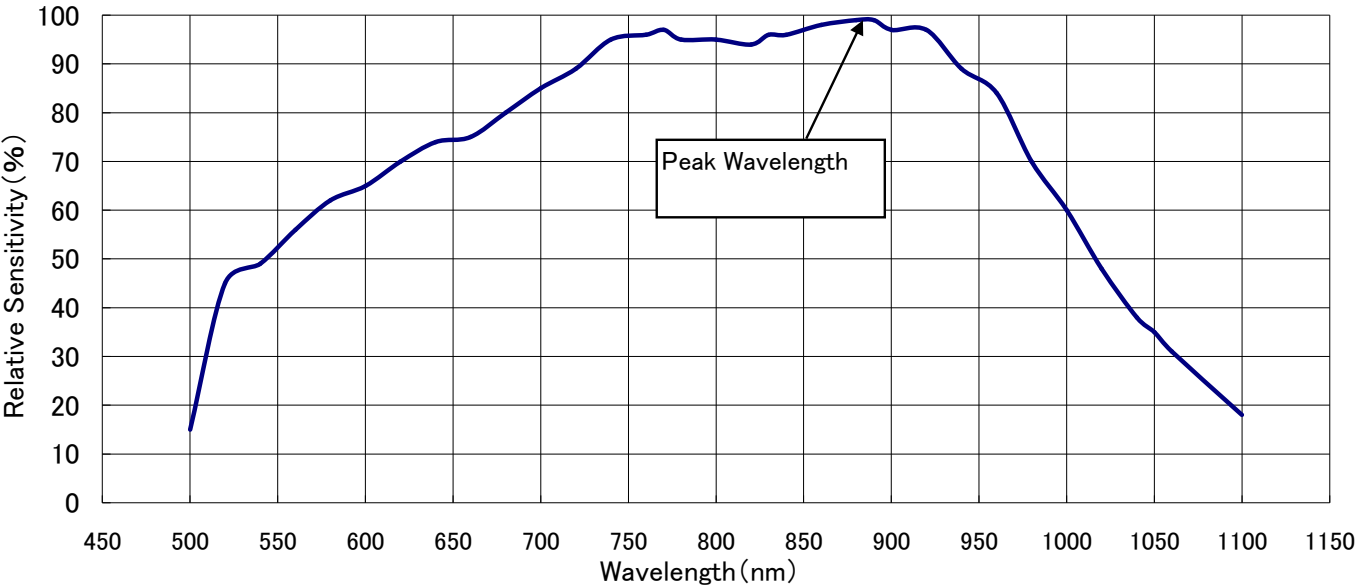
NO.	Parameter	Symbol	condition	SPEC			
				MIN	TYP	MAX	UNIT
1	IV conversion resister (PD1)	RIV1	Iin=10uA	36	44	52	kΩ
2	IV conversion resister (PD2)	RIV2	Iin=10uA	36	44	52	kΩ
3	AMP maximum gain	GM	SENS=5V、f=100kHz	54	57	60	dB
4	AMP minimum gain	GN	SENS=0V、f=100kHz	—	—	0	dB
5	LED emission voltage	VLED	RL=200Ω	2	2.3	2.6	V
6	Stable light level	RANNU	Ratio of stable light level to light level.	115	120	125	%
7	Light level	VNYU		140	180	230	mV
8	MAX shadow level	RSYA	Ratio of MAX shadow level to light level (SENS=5V)	70	75	80	%
9	MIN shadow level		Ratio of MIN shadow level to light level (SENS=0V)	88	93	98	%
10	MAX stable shadow level	RANSY	Ratio of MAX stable shadow level to light level (SENS=5V)	52	58	64	%
11	MIN stable shadow level		Ratio of MIN stable shadow level to light level (SENS=0V)	72	77	82	%
12	Sensitive current	ION	I on=VNYU/(10 ⁻⁶ (GM/20))/RIV1	4.34	6.2	8.06	nA
13	Pulse width of light signal	TTP	An asynchronous mode	2.4	2.7	3.0	us
14	Pulse period of light signal	TTS	nchronous mode	18.0	20.0	22.0	us

【Photodiode characteristics (Reference)】 Ta=25℃

NO.	Parameter	Symbol	condition	MIN	TYP	MAX	UNIT
1	Responsivity	S	780nm、VR=0.7V	0.352	0.440	0.528	A/W
2	Peak Wavelength	λ M	VR=0.7V	—	885	—	nm

(NOTE) This is reference data. It is not data guaranteeing a characteristic.

【Spectral Sensitivity Characteristic (Reference)】 Ta=25℃



1 2 Packaging specification

1 2 – 1 Precautions for handling and storage condition.

(1) When moisture-proof bag is not opened, it is recommended that the products are stored on condition of normal temperature (5~40℃) and normal relative humidity (less than 80%RH).

(2) Precautions at the time of opening the moisture-proof bag.

①When the moisture-proof bag is opened, please be careful to discharge static electricity from workbenches and worker's body.

②After delivery, it is recommended that the devices are used in less than one year in an unopened condition.

(3) Storage condition after opening a moisture-proof bag

①After opening the moisture-proof bag, please store the devices under following conditions.

(A) Temperature : 5~40℃

(B) Relative humidity : less than 60%RH

Please mount the devices within 12 hour after opening moisture-proof bag.

②In case the devices are stored again after opening the moisture-proof bag, please store in the dry box which is installed in the place managed in environment of 12-1(3)① or please perform again moisture-proof packing, and store it. Storage period in the dry box is less than 72 hours.

(4) Baking before mounting

If the storage conditions that is shown in 12-1(3)① are not kept, please perform baking processing on the condition of 120℃ 12 hours.

When the baking processing is performed,

①If this product is a reel state when it was delivered, please bake it as a reel state.(note)

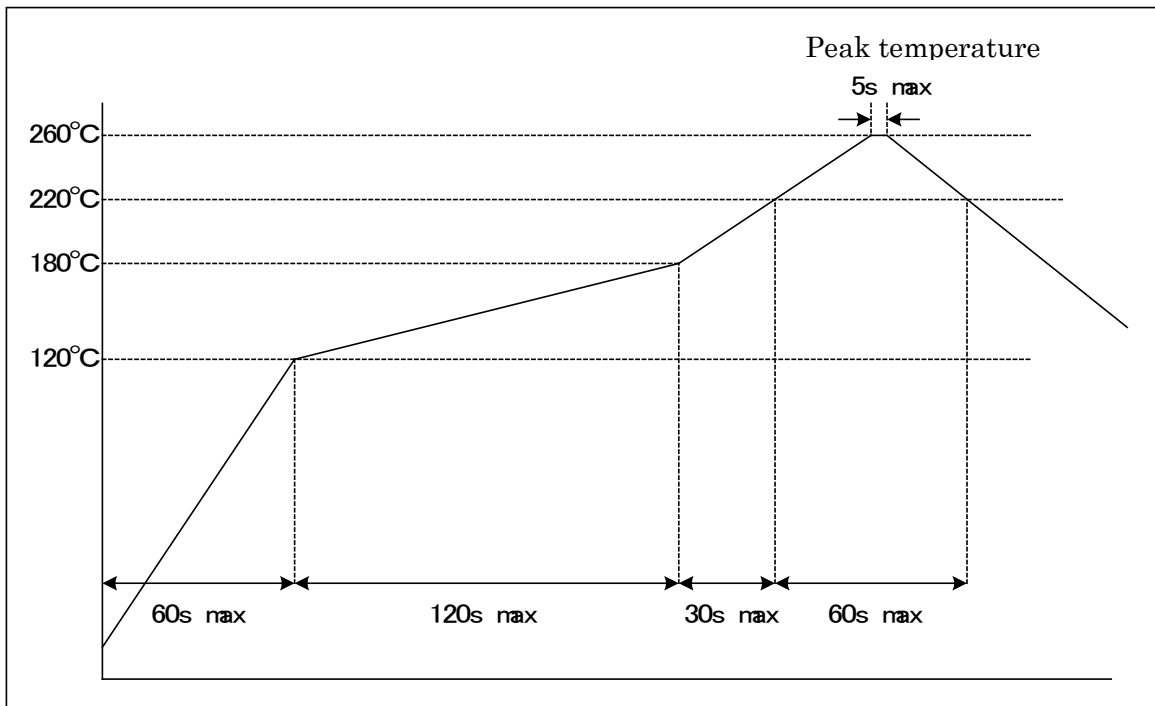
②when the devices is taken out from a reel, please move it to a heat-resistant tray, and bake it.

(note) When warm air bakes the product in a state to be the reel directly, please consider the setting place because reel transformation may occur.

1 2 – 2 Mounting condition

On mounting by the solder, the following reflow conditions and twice reflows are recommended to secure the quality of the device.

- Temperature : 260°C, 5sec MAX (Peak temperature)
120~180°C, 120sec MAX (Preheating)
220~260°C, 60sec MAX
- Temperature measurement point : the surface of device
- Temperature profile : Cf. chart below



1 2 – 3 Attached sentences

- | | |
|---|-------------------------|
| 1) Appearance inspecting standards | ----- (Appended form 1) |
| 2) External dimensions | ----- (Appended form 2) |
| 3) Making specifications | ----- (Appended form 3) |
| 4) Reel and carrier tape specifications | ----- (Appended form 4) |
| 5) Taping specifications 1 | ----- (Appended form 5) |
| 6) Taping specifications 2 | ----- (Appended form 6) |
| 7) Label specifications | ----- (Appended form 7) |
| 8) Packaging specifications | ----- (Appended form 8) |

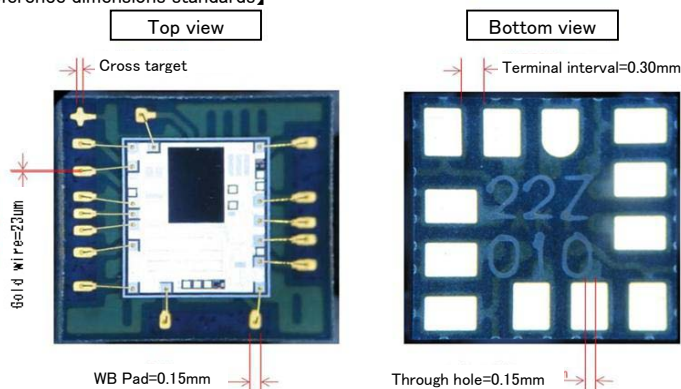
1 2 – 4 The other

If a doubt matter occurs, while discussing it each other in all sincerity, the problem must be solved.

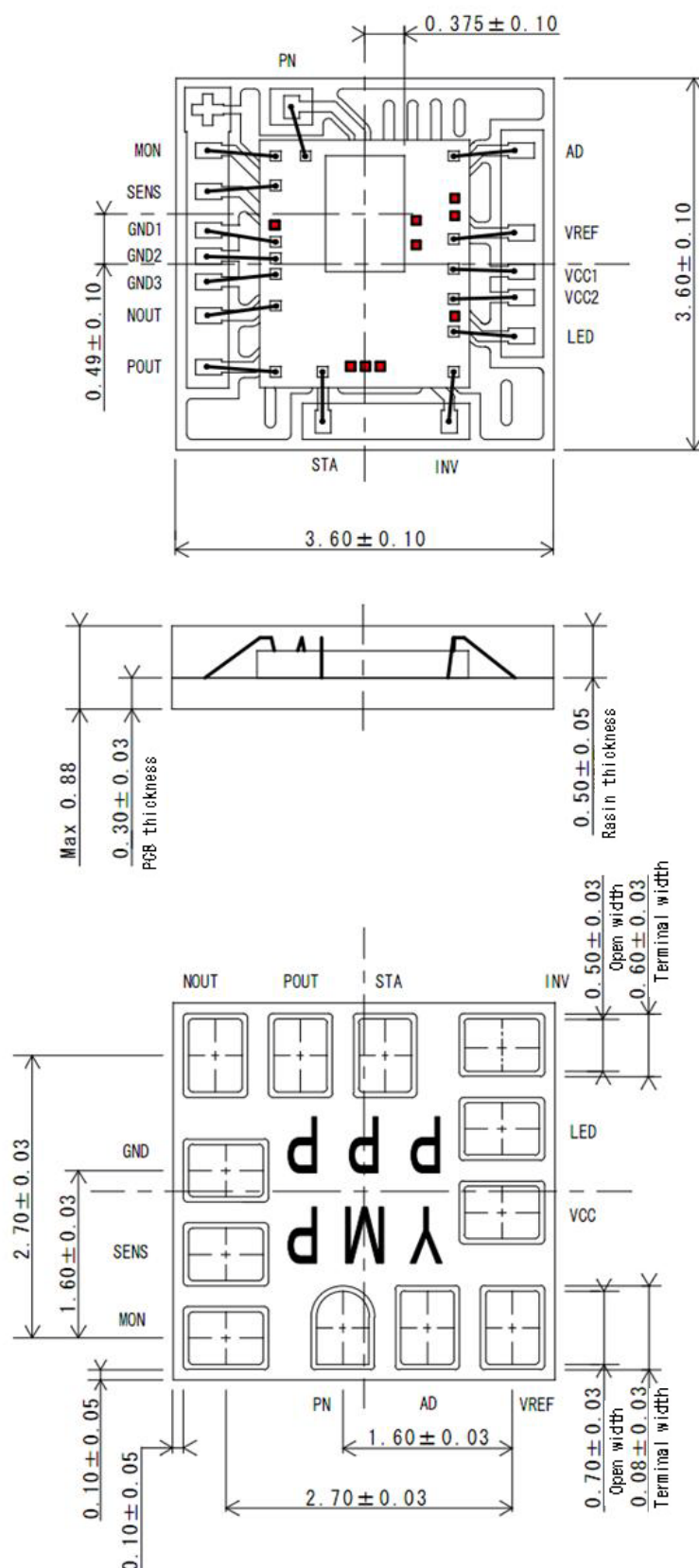
Appearance inspecting standards

Target	Inspection item	Criterion	Fail name
PCB part	1 Crack	1 - 1 At the terminal, the crack is bad.	Crack
		1 - 2 Except the terminal, the crack of $L>0.15\text{mm}$ $W>0.15\text{mm}$ $D>0.15\text{mm}$ is bad.	Crack
	2 Blister	2 - 1 The blister of more than 0.30mm is bad.	Blister
	3 Stain	3 - 1 The stain harming the appearance remarkably is bad.	Stain
	4 Foreign particle	4 - 1 The deposit of more than $\phi 0.30\text{mm}$ that has height and cannot be removed is bad.	Foreign particle
		4 - 2 On the terminal, the foreign particle which cannot be removed is bad.	Foreign particle
	5 Protrusion and Dent	5 - 1 The protrusion or dent on a terminal is bad.	Protrusion, Dent
Mark		5 - 2 Except the terminal, the protrusion and dent of more than $\phi 0.15\text{mm}$ is bad.	Protrusion, Dent
	6 Scratch	6 - 1 The scratch with the depth that base material and copper foil can be identified is bad.	Scratch
	7 Dimension	7 - 1 The PCB which dose not satisfy dimensions specifikasions is bad.	Fail dimension
	8 Mark	8 - 1 The mark which is different from specifikasions is bad.	Fail mark
Rasin par		8 - 2 The device without a mark in the appointed position is bad.	No mark
		8 - 3 The mark that a direction is different from specifikasions is bad.	Reverse mark
		8 - 4 The mark that a letter is blurred, and reading is not possible is bad.	Fail mark
	9 Stain	9 - 1 The stain which cannot be removed by cleaning is bad.	Stain
	# Foreign particle	# - 1 On the photodiode, the foreign particle of more than $\phi 0.10\text{mm}$ is bad.	Foreign particle
		# - 2 Except the photodiode, the foreign particle of more than $\phi 0.15\text{mm}$ is bad.	Foreign particle
	# Scratch	# - 1 The scratch which has the width beyond gold wire width(23um) and has the depth that can be identified is	Scratch
	# Crack	# - 1 Above the photodiode, the crack is bad.	Crack
		# - 2 Except the top surface of the photodiode, the crack of $L>0.15\text{mm}$ $W>0.15\text{mm}$ $D>0.15\text{mm}$ is bad.	Crack
		# - 3 The crack with less than 0.10mm of distance with the gold wire is bad.	Crack
	# Void	# - 1 Above the photodiode, the void of more than $\phi 0.10\text{mm}$ is bad.	Void
		# - 2 Except the top surface of the photodiode, the void of more than $\phi 0.15\text{mm}$ is bad.	Void
		# - 3 The void with less than 0.10mm of distance with the gold wire is bad.	Void
	# Bubble	# - 1 Above the photodiode, the bubble of more than $\phi 0.10\text{mm}$ is bad.	キボウ
		# - 2 Except the top surface of the photodiode, the bubble of more than $\phi 0.15\text{mm}$ is bad.	Bubble
		# - 3 The bubble with less than 0.10mm of distance with the gold wire is bad.	Bubble
		# - 4 When plural bubble occurs, the bubble with more than 0.02mm^2 ($0.15\text{mm} \times 0.15\text{mm}$) of total area is bad.	Bubble
	# Wire exposure	# - 1 When the gold wire is exposed from the resin, it is judged to be bad.	Scratch
	# Chip exposure	# - 1 When the chip is exposed from the resin, it is judge to be bad.	Crack
	# Rasin burr	# - 1 The rasin burr with more than 0.15mm of length is bad. 除去できるものは除去する。	Rasin burr
	# Dimension	# 1 The rasin which dose not satisfy dimensions specifikasions is bad.	Fail dimension
※If the stain and foreign particle can be removed by cleaning, the product are judged to be quality goods. When it is not appointed in particular, IPA is used as the solvent of cleaning.			

【Reference dimensions standards】



Dimensions specifications



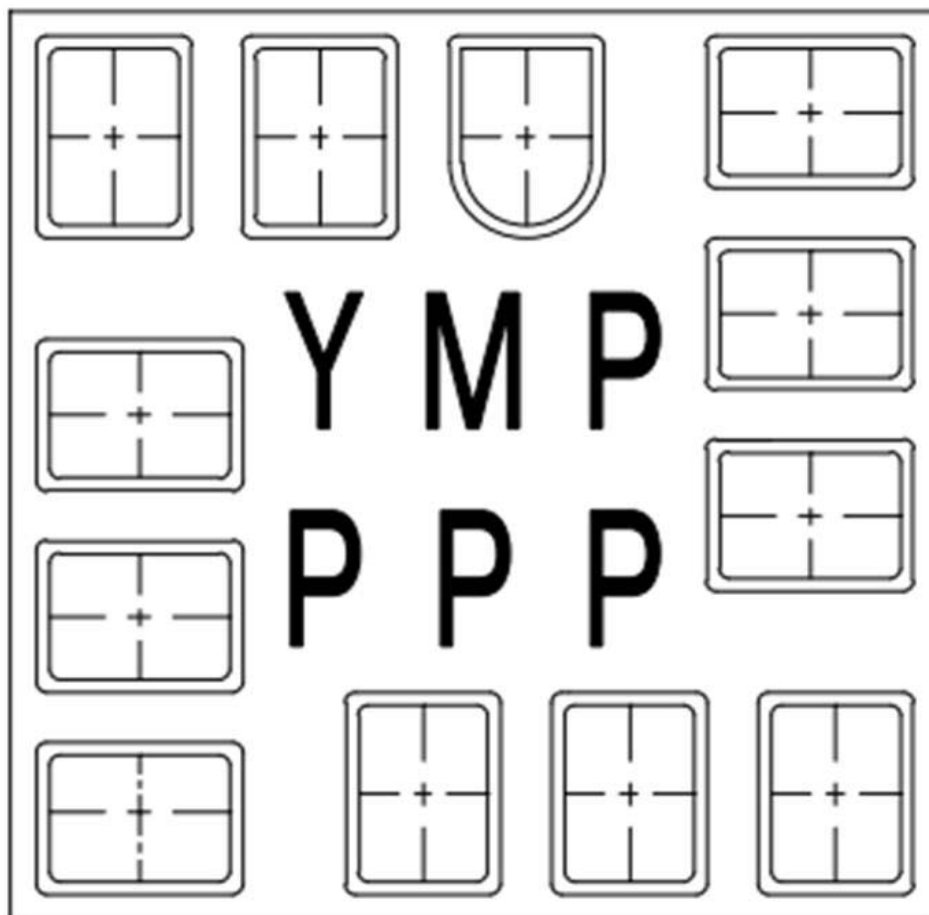
NAME SEP11 Dimensions specifications

NOTE

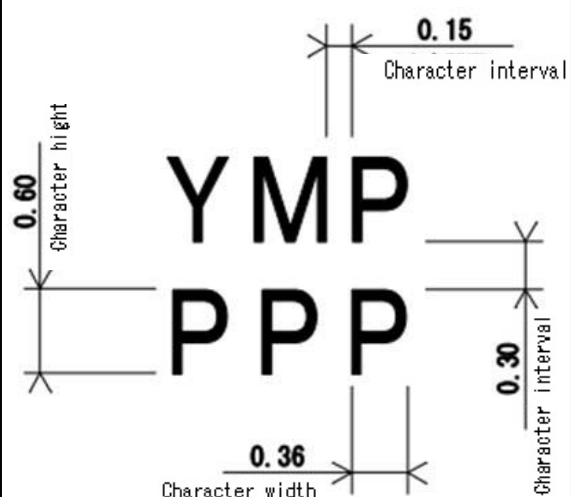
When there is not a mention, the unit is "mm".

No.

Mark specifications



【 Character 】



【 Mark system diagram 】

Y M P
P P P

Y : Year Code
(0、1、2、...、9)

M : Month Code
(1、2、...、9、X、Y、Z)

P : Process Lot No.

NAME SEP11 mark specifications

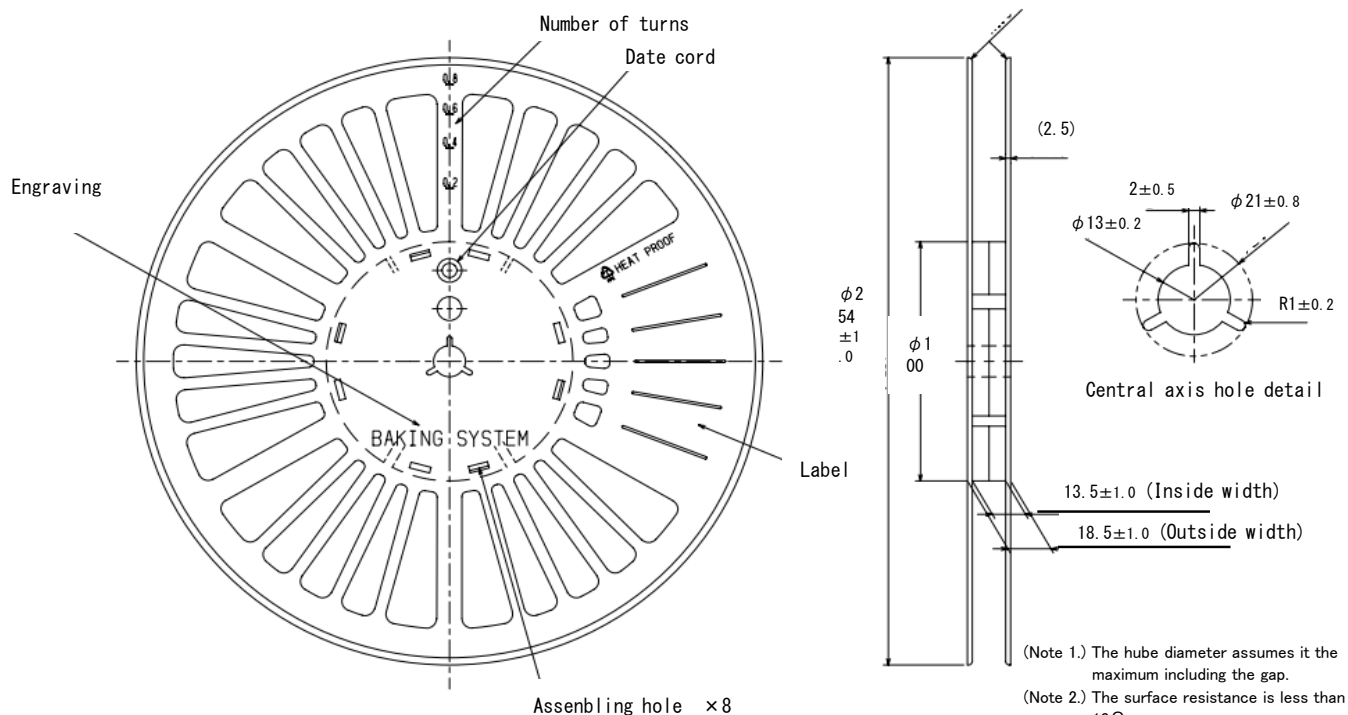
No.

NOTE

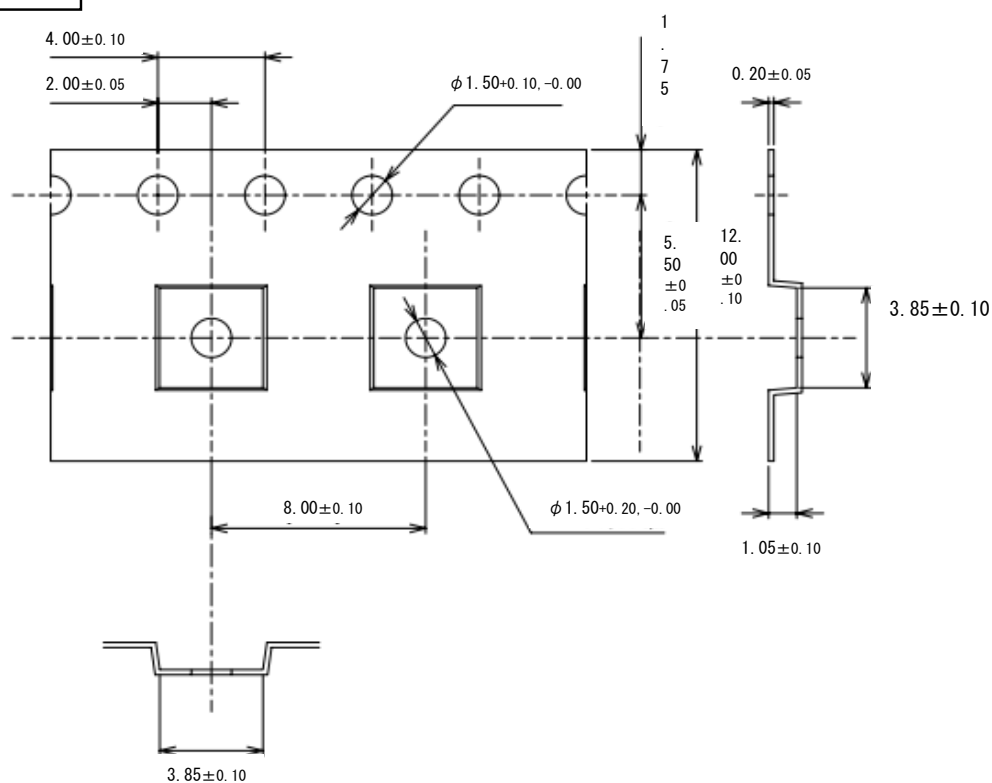
When there is not a mention, the unit is "mm".

Reel and carrier tape specifications

Heat-proof reel



Carrier tape



NAME Reel and carrier tape specifications

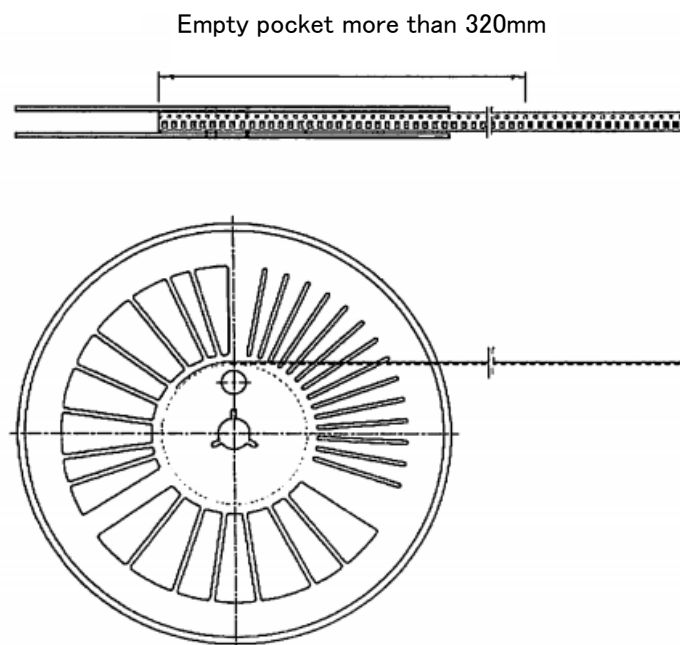
No.

NOTE

When there is not a mention, the unit is "mm".

Taping specifications 1

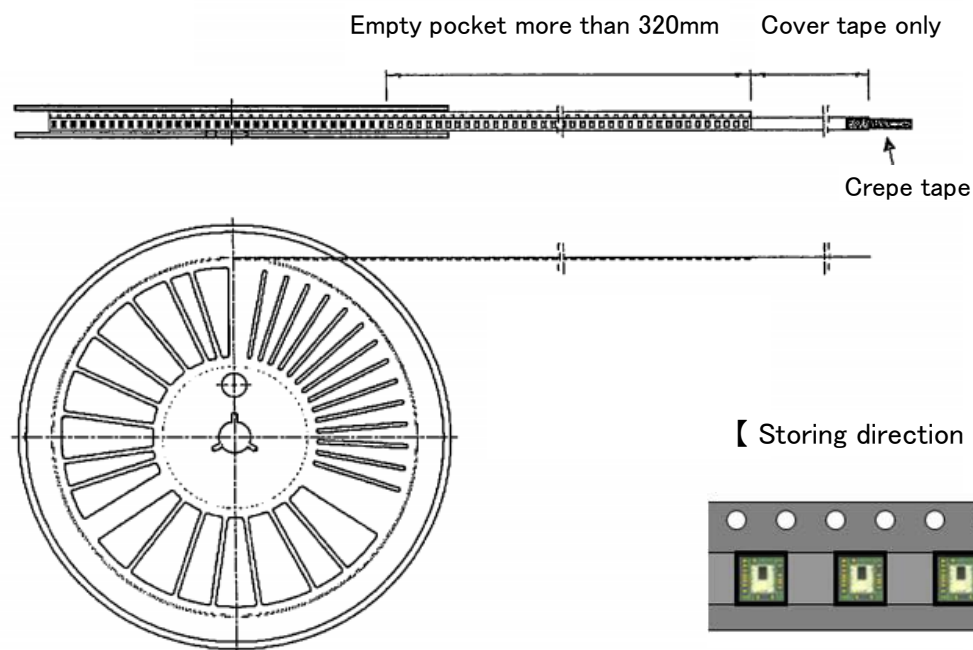
The winding beginning



The 2 pockets are bended and put



The winding end



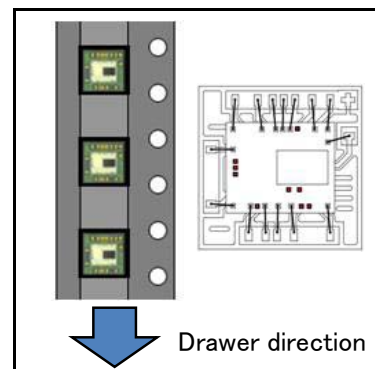
NAME Taping specifications 1

NOTE

No.

Taping specifications 2

- (1) The taping is clockwise twining. When a tape edge is taken out toward, sprocket hole is the right side. (fig-1)
- (2) Please refer to (fig-1) for the device storing direction.
- (3) There are the empty pockets in the winding beginning and the winding end. (fig-2)
 1. The winding beginning more than 320mm
 2. The winding end more than 720mm
 - 2-1 The empty pocket (more than 320mm)
 - 2-2 The cover tape only (more than 400mm)
 3. The end of the cover tape is fixed with crepe tape.

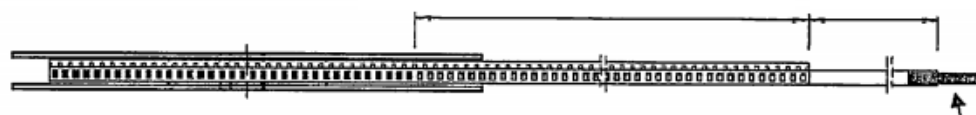


(fig-1)

1. Empty pocket more than 320mm



2-1 Empty pocket more than 320mm 2-2 Cover tape only

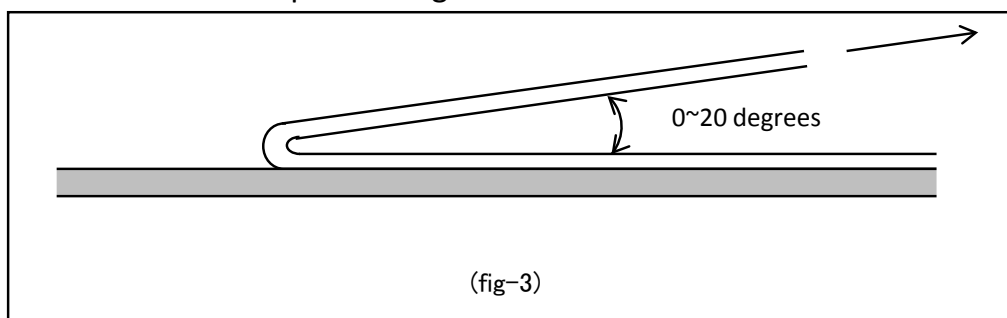


3. Crepe tape

(fig-2)

- (4) The cover tape dose not hang over the sprocket hole.
- (5) There is not the joint in carrier tape and the cover tape.
- (6) The number of the product storing of 1reel is 2,000 normally.
- (7) In case of 0~20 degrees for carrier tape, the peel strength of the cover tape is 0.1~0.7N. (fig-3)

Test method for peel strength



(fig-3)

NAME Taping specifications 2

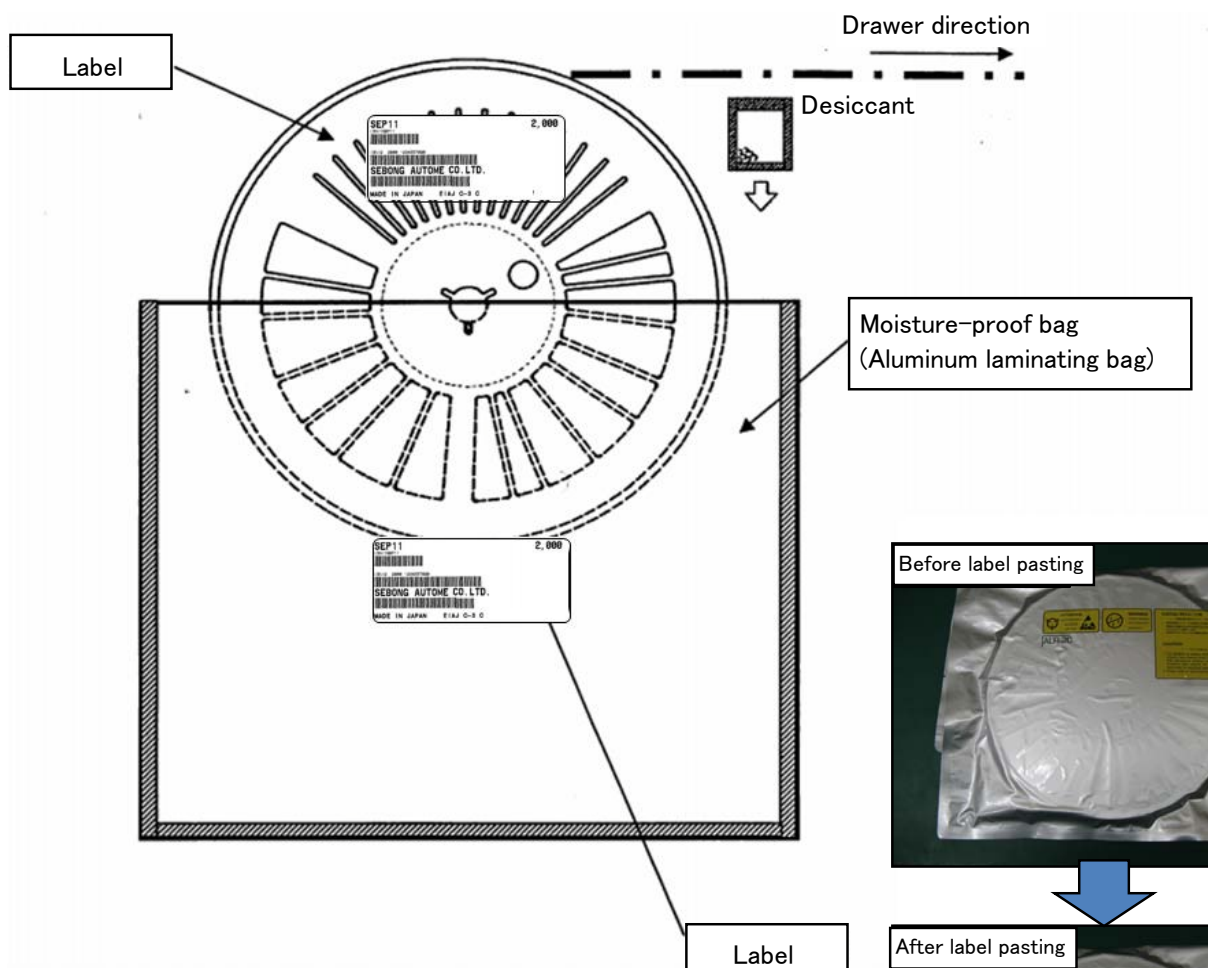
NOTE

No.

Label specifications	
Device name Packing amount and Company name (Half size alphanumeric)	 Label size : 100mm × 40mm
NAME Label specifications	NOTE
NO.	

Packaging specifications

(Label)



NAME	Packaging specifications	NOTE
No.		

Revision record

[illegible]